

**DEPARTMENT OF MECHANICAL ENGINEERING
FACULTY OF ENGINEERING AND ARCHITECTURE
MBM UNIVERSITY JODHPUR**

Subject: 5ME42A(ME) – Manufacturing Technology

Tutorial # 1

Topic: Principal Stress and Strains

1. Considering plane stress system, find magnitudes and directions of principal stresses, maximum shearing stress and corresponding normal stress. Also construct Mohr's circle for
 - a. $\sigma_x = 750 \text{ N/mm}^2$; $\sigma_y = 150 \text{ N/mm}^2$; $\sigma_z = 0$ and $\tau_{xy} = 150 \text{ N/mm}^2$
 - b. $\sigma_x = 450 \text{ N/mm}^2$; $\sigma_y = 450 \text{ N/mm}^2$; $\sigma_z = 0$ and $\tau_{xy} = 150 \text{ N/mm}^2$
2. For problems given in question 1, find yield stress of the material according to:
 - a) Tresca criteria
 - b) Von-Mises Criteria
3. Show through derivation and Mohr's circle that:
 - a. Maximum shear stress is half of the difference between principal stresses
 - b. Normal stress on planes of maximum shear stress is mean of principal stresses
4. Sketch and explain Mohr's circle in three dimensional stresses. Indicate the planes along with the maximum shear stress occur. Show that Von-Mises yield criteria is represented by a circular cylinder in stress space whose axis is equally inclined to the three axes and its radius is $\sqrt{\frac{2}{3}}$ times the yield strength of the metal.

Reference:

(Ch 10 – Principal Stresses and Strains in *Strength of Materials* by Ramamrutham)
(Ch 4, 5 - Principal Stresses and Principal Strains in *Mechanics of Materials* by B C Punmia)
(Ch 3 – Forming Processes in *Manufacturing Sciences* by Ghosh and Mallik)